

Executive Business Report :-

Sales Forecasting and Demand Analysis for Superstore Retail

Executive Summary

In this project, I analyzed four years of Superstore sales data to understand sales trends, predict future sales, identify unusual sales patterns, and group products based on their demand. Different forecasting models were tested, and the best-performing model was selected based on its accuracy. The results can help the company improve inventory planning, reduce unnecessary stock, and make better business decisions.

Key Findings

- Technology was the highest-selling product category during the four-year period.
 - Sales increased during festive and year-end months, showing a clear seasonal pattern.
 - The West and East regions performed better than the other regions.
 - Average shipping time remained almost the same across all regions, showing a stable delivery process.
 - The sales trend showed steady growth over time with some seasonal fluctuations.
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Three-Month Sales Forecast

The selected forecasting model predicts that sales will remain stable and are likely to grow slightly over the next three months. Although actual sales may change because of promotions or market conditions, the overall business outlook is positive and no major decline is expected.

Top Anomalies Found

- A very high sales spike was observed during the festive season, most likely due to promotional offers and increased customer demand.
 - A sudden drop in sales may have occurred because of lower demand or temporary stock shortages.
 - Another unusual increase in weekly sales could be linked to bulk customer orders or a special sales campaign.
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Product Demand Segmentation

Products were divided into four demand groups using K-Means clustering.

- High Volume, Stable Demand: Maintain sufficient inventory because these products sell consistently.
 - Growing Demand: Increase stock gradually to meet future demand.
 - Low Volume, High Volatility: Keep limited inventory and monitor demand regularly.
 - Declining Demand: Reduce stock levels and clear existing inventory through discounts if required.
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Business Recommendations

1. Maintain higher stock levels for Technology products because they generate the highest sales.
 2. Increase inventory before festive seasons to avoid stock shortages during high demand.
 3. Use demand-based inventory planning so that fast-selling products remain available while slow-moving products do not occupy unnecessary warehouse space.
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Risk and Limitation

The forecasting results are based on historical sales data. Unexpected events such as market changes, economic conditions, or supply chain disruptions may affect future sales. Therefore, the forecasting model should be updated regularly with new data.

Conclusion

The analysis shows that sales forecasting and demand segmentation can support better inventory planning and business decisions. By following these recommendations, the company can improve product availability, reduce inventory costs, and manage future demand more effectively.